

# Oak Ridge In Hot Water

Scientists at Oak Ridge National Laboratory (ORNL) are taking on the second biggest energy user in the home—hot water. One way to minimize water-heating energy is to combine this technology with other energy-using devices. Two of ORNL's current projects take this approach. One project uses the heat given off by refrigerators to help heat water and also to cool the house in the summer. The other project combines a water heater with a dehumidifier. A third project is aimed at effectively marketing new technologies, such as ORNL's heat pump water heater (HPWH). Finally, a project has studied the efficiencies of hot water distribution systems and has made recommendations for improvements.

## Heating Water with Waste Heat

Cleaning the coils on the back of your refrigerator won't save you any energy (see "Common Energy Myths," *HE* Sept/Oct '01 p. 33). However, researchers at ORNL, working together with Habitat for Humanity, have designed a home in the Harmony Heights subdivision in Lenoir City, Tennessee, that makes use of the heat from those refrigerator coils to heat water and cool a house. The near-net-zero energy house includes a HPWH, donated by Enviromaster International LLC, that is integrated with the refrigerator as well as with the unvented crawlspace and the heating and cooling system (see "Near-Net-Zero Energy").



The E-tech R106 HPWH mounts on top of a standard water heater.

Waste heat is used to preheat water in the HPWH. When the home is in cooling mode, waste heat from the refrigerator is directed through motorized dampers to the HPWH situated in a closet behind the refrigerator. The HPWH uses this waste heat and then returns cool exhaust air into the kitchen to lessen the load on the air conditioner. When the heat is on, a fan sends crawlspace air to the HPWH, which then exhausts it to the outside. (This does not take heat from the house and transfer it to the HPWH. There is more heat transfer from the ground to the outside air that is drawn into the crawlspace than there is heat transfer from the house to the crawlspace, since the floor is well insulated.) Because the HPWH uses multiple sources of heat, the HPWH

coefficient of performance (COP) has ranged from 2.0 to 2.2 during a year of testing.

## Water-Heating Dehumidifier

As new, well-insulated homes become tighter, potential moisture problems arise, especially in homes that bring in humid outside air for fresh air. Humid air can promote the growth of mold, leading to poor indoor air quality and rot. The water-heating dehumidifier (WHD) in development at ORNL is meant to address humidity control. Several configurations will be modeled and tested to identify the most marketable, least expensive, and most energy-efficient device.

The WHD will include a compressor, two condensers, an evaporator, and an expansion device. One condenser will be used when the WHD is operating as a water heater. When the tank is full of hot water, the WHD changes over to operate as a dehumidifier. In this mode, the second, air-cooled condenser becomes operational. The goal is to create a WHD with an energy factor of 1.2 that will remove 10–24 liters of water from the air each day. ORNL estimates that the WHD will cost \$400–\$600. This price is comparable to that of a 65-pint dehumidifier that costs approximately \$250 and that of a 50-gallon electric water heater that costs \$200–\$300.

The WHD will be prototyped through a collaborative effort among ORNL, the Education and Research

## Near-Net-Zero Energy

The near-net-zero energy house in Lenoir City, Tennessee, where ORNL is testing its HPWH incorporates structural insulated panels donated by the Structural Insulating Panel Association for the floors, walls, and roof, as well as a raised-metal-seam roof provided by the Metal Roof Alliance. The solar collectors on this roof

are mounted with clips, to avoid penetrating the metal. The Tennessee Valley Authority (TVA) provided a grid-connected, 2 kW PV system, designed and installed by Big Frog Mountain of Chattanooga, Tennessee. Under its Green Power Switch Generation Partner program, TVA, along with the Lenoir City Utilities Board, will pay the homeowner

\$500 plus \$0.15/kWh for all the alternating current (AC) solar power generated by this system for at least ten years. The buyback offer is 2.4 times the residential retail rate for electricity. Andersen Windows provided high-efficiency windows, and Du Pont installed them and also donated the weather barrier system.

Consortium of the Western Carolinas, Western Carolina University, Asheville–Buncombe Technical Community College, the American Carolina Stamping Company, and Clemson University. The WHD will be marketed by the newly created Western North Carolina Center for Technology Development (a partnership among ORNL, Institute at Biltmore, University of North Carolina at Asheville, Clemson University, and Western Carolina University).

Ten or more of the new WHD will be field tested in North Carolina and Tennessee homes in 2005.

### *Heat Pump Water Heater*

ORNL and Arthur D. Little helped engineers at ECR International, Incorporated, to create a HPWH that could be installed on top of a conventional electric water heater. The drop-in unit is 3 times as energy efficient as the most efficient conventional electric water heater. Since the HPWH draws heat from the surrounding air, it also has the capacity to cool and dehumidify the space where it is installed. (For more on the HPWH, see “All Pumped Up,” *HE* Nov/Dec ’02, p. 30.)

Because of the high first cost, and past serviceability issues, the HPWH has made little impact on the water heater market, despite its energy efficiency. ORNL is conducting market research with the hope of integrating what they learn into their research and development efforts. Their goal is to work with industry partners to better market energy-efficient technologies such as the HPWH.

### *Hot Water Distribution Systems*

ORNL was commissioned by the California Energy Commission to study the impact of various hot water distribution systems on a house’s energy use, water use, and the wait time for hot water to reach the end user. To this end, ORNL created a model to study various configurations of housing type and hot water distribution system.

For a complete description of the study and its outcomes and recommendations, see “Are You Getting into Hot Water?” *HE* Sept/Oct ’03, p. 33, and “Hey, Where’s the Hot Water?” *HE* Sept/Oct ’04, p. 36.

—**Jim Gunshinan** 

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#### **For more information:**

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